Jacob Mitchell Springer

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Education

Ph.D., Machine Learning Aug 2022 – Present
Carnegie Mellon University, Pittsburgh, PA
Advisor: Aditi Raghunathan. Supported by the NSF Graduate Research Fellowship.

B.A., Mathematics and Computer Science Aug 2017 – May 2022
Swarthmore College, Swarthmore, PA

Publications

† Co-last authors.

19. C. Li, J. Gai, **J. Springer**†, A. Raghunathan†. *Pretraining Data Scale Reverses the AdamW–Muon Fine-Tuning Gap*. Workshop on Methods and Opportunities at Small Scale (MOSS) @ COLM, 2026.
18. L. Feng, G. R. Ghosal, **J. Springer**, Z. Zhong, A. Raghunathan. *Early Data Exposure Improves Robustness to Subsequent Fine-Tuning*. COLM, 2026.
17. **J. Springer**, M. Advani, L. Aichberger, A. Bradley, E. Malach, O. Saremi, S. Williamson, P. Nakkiran, E. Littwin, A. Raghunathan. *Annotations Mitigate Post-Training Mode Collapse*. ICML, 2026.
16. I. Watts, C. Li, S. Goyal, **J. Springer**†, A. Raghunathan†. *Sharpness-Aware Pretraining Mitigates Catastrophic Forgetting*. ICML, 2026. **Oral, ICBINB Workshop @ ICLR 2026**
15. A. Kulkarni, **J. Springer**, A. Subramonian, S. Swayamdipta. *Disentangling Geometry, Performance, and Training in Language Models*. ICML, 2026. **Spotlight**
14. **J. Springer**, S. Goyal, K. Wen, T. Kumar, X. Yue, S. Malladi, G. Neubig, A. Raghunathan. *Overtrained Language Models Are Harder to Fine-Tune*. ICML, 2025. **Outstanding Paper, SCOPE Workshop @ ICLR 2025; Entropic Paper Award, ICBINB Workshop @ ICLR 2025**
13. **J. Springer**, S. Kotha, D. Fried, G. Neubig, A. Raghunathan. *Repetition Improves Language Model Embeddings*. ICLR, 2025.
12. T. Kim, **J. Springer**, A. Raghunathan, M. Sap. *Mitigating Bias in RAG: Controlling the Embedder*. ACL Findings, 2025.
11. **J. Springer**, V. Adlakha, S. Reddy, A. Raghunathan, M. Mosbach. *Understanding the Influence of Synthetic Data for Text Embedders*. ACL Findings, 2025.
10. **J. Springer**, V. Nagarajan, A. Raghunathan. *Sharpness-Aware Minimization Enhances Feature Quality via Balanced Learning*. ICLR, 2024.
9. S. Kotha, **J. Springer**, A. Raghunathan. *Understanding Catastrophic Forgetting in Language Models via Implicit Inference*. ICLR, 2024.
8. H. T. Jones, **J. Springer**, G. T. Kenyon, J. Moore. *If You've Trained One You've Trained Them All: Inter-Architecture Similarity Increases With Robustness*. UAI, 2022. **Oral presentation**

7. **J. Springer**, M. Mitchell, G. T. Kenyon. *A Little Robustness Goes a Long Way: Leveraging Robust Features for Targeted Transfer Attacks*. NeurIPS, 2021.
6. **J. Springer**, B. M. Reinstadler, U.-M. O'Reilly. *STRATA: Simple, Gradient-Free Attacks for Models of Code*. Workshop on Adversarial Learning Methods @ KDD, 2021.
5. **J. Springer**, G. T. Kenyon. *It's Hard for Neural Networks to Learn the Game of Life*. IJCNN, 2021.
4. **J. Springer**, M. Mitchell, G. T. Kenyon. *Adversarial Perturbations Are Not So Weird: Entanglement of Robust and Non-Robust Features in Neural Network Classifiers*. Preprint, 2021.
3. D. A. Wang, C. M. S. Strauss, **J. Springer**, A. Thresher, H. Pritchard, G. T. Kenyon. *Sparse MP4*. IEEE SSIAI, 2020.
2. **J. Springer**, C. S. Strauss, A. M. Thresher, E. Kim, G. T. Kenyon. *Classifiers Based on Deep Sparse Coding Architectures Are Robust to Deep Learning Transferable Examples*. Preprint, 2018.
1. **J. Springer**, W. Feng. *Teaching with angr: A Symbolic Execution Curriculum and CTF*. USENIX Workshop on Advances in Security Education, 2018.

Research Experience

Research Intern Jun – Sep 2025
Apple Machine Learning Research, Cupertino, CA
Advised by Etai Littwin.

Graduate Research Assistant Aug 2022 – Present
Carnegie Mellon University, Pittsburgh, PA
Advised by Aditi Raghunathan.

Research Assistant, ML & Computational Neuroscience Jan – Jul 2022
Cold Spring Harbor Laboratory, Cold Spring Harbor, NY
Advised by Anthony Zador.

Research Assistant, ML & Computational Neuroscience Jun 2018 – Dec 2021
Los Alamos National Laboratory, Los Alamos, NM
Advised by Garrett Kenyon.

Research Intern Jun – Aug 2020
MIT, Cambridge, MA
Advised by Una-May O'Reilly.

Research Intern, Computer Security Education Jun – Aug 2017
Portland State University, Portland, OR
Advised by Wu-chang Feng.

Invited Talks

- *Don't (Just) Minimize Your Pretraining Loss.* ML Foundations Seminar, Kempner Institute, Harvard University, Apr 2026.
- *Overtrained Language Models Are Harder to Fine-Tune.* Waymo, Aug 2025.
- *Echo Embeddings & Overtrained Language Models Are Harder to Fine-Tune.* Summer of Data Seminar, DatologyAI, Jun 2025.
- *Overtrained Language Models Are Harder to Fine-Tune.* Translate Reading Group, Google, Jun 2025.

- *Overtrained Language Models Are Harder to Fine-Tune.* FLAME Center Seminar, Carnegie Mellon University, Apr 2025.
- *Repetition Improves Language Model Embeddings.* Foundation and Language Model (FLAME) Seminar, Carnegie Mellon University, Mar 2024.
- *What Can Adversarial Examples Tell Us About Similarities Between Neural Networks?* Pacific Northwest Seminar on Topology, Algebra, and Geometry in Data Science, University of Washington, Feb 2023.

Teaching

- **Teaching Assistant**, 10-715: Advanced Introduction to Machine Learning. Instructor: Nihar B. Shah. Carnegie Mellon University, Fall 2025.
- **Teaching Assistant**, 15-789: Theoretical and Empirical Foundations of Modern Machine Learning. Instructor: Aditi Raghunathan. Carnegie Mellon University, Fall 2024.

Awards

Outstanding Paper, SCOPE Workshop @ ICLR (for "Overtrained Language Models Are Harder to Fine-Tune")	2025
Entropic Paper Award, ICBINB Workshop @ ICLR (for "Overtrained Language Models Are Harder to Fine-Tune")	2025
Hertz Fellowship, Finalist	2023
NSF Graduate Research Fellowship	2022
Barry M. Goldwater Scholarship	2020
National Merit Scholarship, Finalist	2017